Pandas Daily Practice

A 15-25 minute retrieval routine for making labeled data analysis stick

The goal is not to memorize every method. Build one reliable mental pipeline: **inspect, select, transform, preserve row identity, aggregate, verify.**

The daily routine

Minutes	Activity	What to do
0-3	Retrieve	Write yesterday's pandas patterns from memory. Do not open notes yet.
3-6	Predict	Before each line, predict: DataFrame, Series, scalar, or index label? Also predict shape.
6-18	Build	Complete one small pipeline on a familiar or fresh table. Run after each meaningful step.
18-22	Verify	Check one result by hand and inspect complete rows, not isolated columns.
22-25	Reflect	Write one sentence: what returned an unexpected object or shape today?

Core retrieval sheet

Purpose	Pattern
One column -> Series	`s = df["Exam1"]`
Several columns -> DataFrame	`table = df[["Exam1", "Exam2"]]`
Derived column	`df["StudentMean"] = df[exam_cols].mean(axis=1)`
Complete-row filter	`selected = df.loc[df["StudentMean"] >= 85]`
Maximum row label	`label = df["StudentMean"].idxmax()`
Complete maximum row	`top_row = df.loc[label]`
Grouped summaries	`summary = df.groupby("Band")["StudentMean"].agg(["count", "mean"])`
Any missing exam value	`df[exam_cols].isna().any().any()`

Memory rule: **loc uses labels; iloc uses integer positions; idxmax returns a label.**

Seven-day practice cycle

Day 1 - Rebuild the exam report

Without notes, load exams.csv, inspect it, derive StudentMean, recover the top complete row, filter qualifying rows, assign bands, and group by band.

Day 2 - Object and shape predictions

Use the exam table. Before every selection or reduction, write the expected object type and shape. Confirm each prediction after running.

Day 3 - Product revenue

Create Product, Category, Price, and UnitsSold. Derive Revenue, recover the maximum-revenue row, filter above a threshold, and group by Category.

Day 4 - Debugging day

Intentionally use one wrong axis, one extra pair of brackets around a mask, and iloc with an idxmax label. Explain each failure, then repair it.

Day 5 - Validation retrieval

On a small CSV, check exact columns, empty data, missing values, numeric conversion, duplicates, and a valid numeric range.

Day 6 - Chaining practice

Write groupby and aggregation first as named intermediate variables. Then rewrite the same logic as one readable chain.

Day 7 - Fresh mini-challenge

Use a new table and complete the pipeline without looking at examples. End with a two-sentence explanation of row identity and grouping.

Fresh mini-challenge specification

Create a small CSV with these columns: **Product, Category, Price, UnitsSold**. Use at least six rows and two categories. Then complete the following without a Python row loop:

- [ ] Load the file and inspect shape, columns, and dtypes.
- [ ] Require the exact expected schema and reject missing values.
- [ ] Create Revenue as Price multiplied by UnitsSold.
- [ ] Recover the complete maximum-revenue row without sorting.
- [ ] Filter complete rows whose Revenue meets a threshold you choose.
- [ ] Create Low, Medium, and High revenue bands using Boolean masks and loc.
- [ ] Group by Category and calculate product count and mean Revenue.
- [ ] Verify one Revenue value and one group mean by hand.

Daily prediction prompts

- [ ] What object will this expression return: DataFrame, Series, scalar, or label?
- [ ] What shape should the result have?
- [ ] Does this operation preserve complete rows and their index labels?
- [ ] Which axis is being reduced, and what remains afterward?
- [ ] Can I verify one value manually before trusting the whole result?

End-of-session log

Date	Dataset / task	One thing recalled	One shaky point

Success criterion

You are ready to move on when you can complete the mini-challenge while correctly predicting the main object types and recovering complete rows without losing identity.